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Time-Series Few Shot Anomaly Detection for HVAC Systems

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Abstract:
One of the main challenges in data-driven anomaly detection of building heating, ventilation, and air conditioning systems is the limited amount of labeled data available for training deep learning algorithms. Therefore, developing methods that reuse available data across different systems is the main motivation for this work. We present a few-shot domain adaptation method based on a long short-term memory (LSTM) autoencoder (AE) neural network learning scheme. Our method only requires nominal instances from the source domain and a few nominal instances from the target domain. Results show that our method outperforms existing unsupervised approaches for different fault scenarios.

Keywords: Anomaly Detection, Building Systems, Domain Adaptation, Few-Shot Learning, Machine Learning Methods

1. INTRODUCTION

Buildings are a major contributor to global energy consumption and greenhouse gas (GHG) emissions, accounting for approximately 33% of the energy use and 37% of GHG emissions (Rumsey, 2022). Nearly 75% of energy consumed in buildings is utilized for hot water and space heating, ventilation, and air conditioning (HVAC) systems (REN21, 2023). Faults in HVAC systems are a notable concern, with studies indicating that between 15% and 30% of energy can be wasted due to system faults or improper control of operations (Katipamula and Brambley, 2005). Therefore, anomaly detection and diagnostics are important for the reliability of system operations and for overall energy sustainability.

Anomaly detection is the process of identifying anomalous system behaviors and unexpected events that deviate from normal operating modes. A common problem in HVAC system anomaly detection is the limited amount of labeled data available when deploying a new building (target domain) (Pang et al., 2021). To address this issue, data from other similar buildings (source domains) can be leveraged to improve the anomaly detection performance in the target domain. Examples of source domains include data from the same building from previous years, simulated building anomalies, and data from similar buildings. This area of research is called *domain adaptation* (DA)

Domain adaptation-based anomaly detection methods have several variants based on different assumptions about the data availability in the source and target domains. Supervised DA methods assume the availability of labeled data from both normal and anomalous conditions in the source domain, but only labeled normal data in the target domain. Semi-supervised DA methods also assume the availability of labeled data from both normal and anomalous conditions in the source domain, but only a few labeled examples (either normal or anomalous) in the target domain. Unsupervised DA methods do not require labeled data in either the source or target domains but usually need a large amount of data from multiple source domains for training.

However, these DA methods have limitations when applied to HVAC anomaly detection. Supervised DA methods require a significant amount of labeled anomalous data in the source domain, which can be difficult and expensive to obtain in real-world HVAC systems. Semi-supervised DA methods alleviate some of these issues by requiring only a few labeled examples in the target domain. However, they still assume the availability of labeled anomalous data in the source domain, and the performance of semi-supervised DA methods heavily relies on the quality and diversity of the few labeled examples in the target domain. Unsupervised DA methods eliminate the need for labeled data in both the source and target domains. However, they often require a large amount of data from multiple source domains to learn a robust representation of normal behavior. Additionally, unsupervised DA methods often suffer from high false alarm rates and low accuracy, as they lack the guidance of labeled anomalies during training.

Inspired by *few-shot learning*, which addresses the problem of low accuracy of unsupervised DA methods and the high labeling cost of supervised DA methods, this paper

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proposes a few-shot domain adaptation model built on a Long-Short Term Memory (LSTM)-based Autoencoders (AE) neural network architecture, which requires nominal samples from a single source domain and a few nominal samples from the target domain. Our proposed method first trains an anomaly detection classifier on the source domain and then fine-tunes it on a limited number of nominal samples from the target domain, enabling the reuse of previous knowledge to reduce training complexity and increase accuracy. We demonstrate satisfactory performance based on a realistic case study created for fault detection analysis by the Lawrence Berkeley National Laboratory (LBNL) in the USA.

This paper is organized as follows. Section 2 introduces related work in time-series anomaly detection. Section 3 presents the details of our proposed method. Section 4 illustrates the experimental setup and analysis of results. Finally, Section 5 concludes the article and highlights future research directions.

2. RELATED WORK

Anomaly detection in building datasets that are partially or completely unlabeled is traditionally solved using unsupervised methods (Gunay and Shi, 2020; Lei et al., 2023) or residual-based (Himeur et al., 2021). Many buildings applications use autoencoders (Blázquez-García et al., 2021) or LSTM-based autoencoders (Wei et al., 2023; Park et al., 2023; Coursey et al., 2023). LSTMs capture temporal relationships in the time-series data. However, these methods are confined to single-domain applications and are not adaptable to tasks for a different building, in which case we may need to re-extract the features and retrain a model.

To reduce the complexity and effort to label or develop models for each new dataset, transfer learning and DA have been proposed to solve a task in a target domain using knowledge from a related source domain (Motiian et al., 2017; Achituve et al., 2021; Xu et al., 2019; Kumagai et al., 2019). Although these methods generalize well to new domains, they assume that anomalous and nominal labels are available in the source or target domains. This can become problematic in real applications due to the rarity of anomalous events. Andrews et al. (2016) propose a transfer anomaly detection method that utilizes a one-class support vector machine to estimate the support of the high-dimensional distribution for the target domain, but they assume all normal instances to be available for training. This setting might be inapplicable for large building datasets where only a few normal instances are available.

Few-shot learning has emerged as a solution for reducing the required volume of normal instances from the target dataset (Andrychowicz et al., 2016; Belton et al., 2023), providing the advantages of low complexity given its low data requirements and lower training time. Our proposed method leverages the commonly used LSTM AE architecture and combine it with few-shot learning to improve the performance and cut down the complexity of anomaly detection in partially labeled building datasets. With few-shot domain adaptation, we only need normal samples from the source domain and a few normal instances from the target domain.

Table 1. Example of Best LSTM-AE Model Architecture Obtained from Random Search

Layer (type)	Hyperparameters
Input	Shape=[Samples, Timesteps, Features]
LSTM	OutputDim=8
Dropout	20%
LSTM	OutputDim=4
Dropout	20%
RepeatVector	OutputDim=4
LSTM	OutputDim=4
Dropout	20%
LSTM	OutputDim=8
Dropout	20%
TimeDistributed	
Output	Shape=[Samples, Timesteps, Features]

3. FEW-SHOT LSTM AUTOENCODER

This section introduces our proposed model that combines an LSTM autoencoder (AE) with few-shot fine-tuning to detect anomalies in multivariate time-series data. We start by training an LSTM-AE on all the nominal data – instances without faults or outliers – from the source domain dataset. The model then undergoes fine-tuning with a small set of k nominal instances from the target domain, which may represent data from a different year or building. This is because only a small amount of data may be available from the target domain. This step updates the weights of the LSTM-AE to better accommodate the specific characteristics and patterns of the target dataset.

The fine-tuning step partially addresses the distributional shift in the data. One relevant assumption for the source and target datasets is that the relations among variables hold to some extent, which is a reasonable assumption for real-life HVAC systems. By learning from a few examples of nominal instances from the target set, the fine-tuned model adjusts its learned distribution to align with that of the target domain. An overview of the process of our approach is shown in Algorithm 3.1.

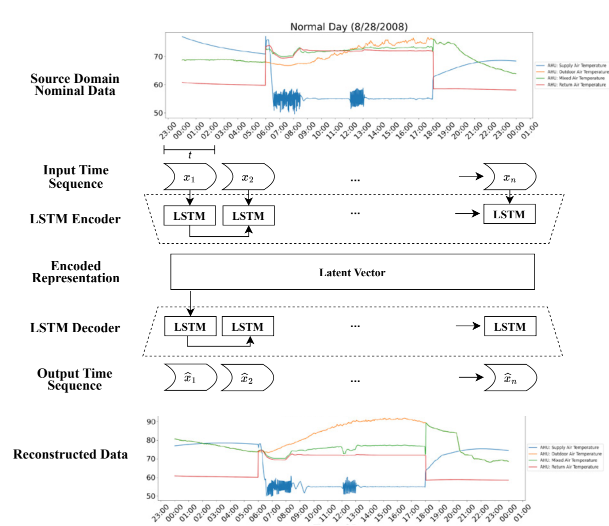

Fig. 1. LSTM autoencoder architecture

3.1 LSTM-Autoencoder

During data preprocessing stage, we append the preceding *Timesteps* of samples to each data point of source and target datasets so that they capture the spatio-temporal relationships as the input to an LSTM. All nominal time sequences from the source dataset are used for training. The target dataset is divided into two sets, one composed solely of nominal time-sequence data that is used for subsequent model fine-tuning, and the second, composed of both nominal and anomalous samples that is reserved for testing.

Figure 1 shows the overall architecture of our LSTM-AE. The LSTM-AE is trained to minimize the mean squared error (MSE) between the input and its reconstructed representation. An outline of our best model architecture obtained based on a random search method for hyperparameter tuning is shown in Table 1.

Algorithm 1 Few-Shot LSTM-AE Anomaly Detection

Data:
Source domain nominal samples $\{x_0, x_1, \ldots, x_n\}$,
k shots of target domain nominal samples $\{x'_0, x'_1, \ldots, x'_k\}$,
Target domain testing set $\{x''_0, x''_1, \ldots, x''_m\}$,
Result:
An anomaly classifier trained on source domain, fine-tuned on k nominal samples of the target domain, predicting anomaly/fault labels for testing data

`/* Phase 1: Data preprocessing */`
X_i, X'_i, X''_i: sets of input based on timesteps ($t = 10$)
$X_i = \{x_i, \ldots, x_{i+t}\}$
$X'_i = \{x'_i, \ldots, x'_{i+t}\}$
$X''_i = \{x''_i, \ldots, x''_{i+t}\}$

`/* Phase 2: LSTM-AE training */`
for $X_i \in \{X_0, X_1, \ldots, X_n\}$ **do**
    $\hat{X}_i = Reconstructed(X_i)$
    $L_{\text{MSE}}[i] = \sum(X_i - \hat{X}_i)^2$
    Update LSTM-AE to minimize L_{MSE}
end

`/* Phase 3: LSTM-AE few-shot fine-tuning */`
for $X'_i \in \{X'_0, X'_1, \ldots, X'_n\}$ **do**
    $X'_i = Reconstructed(X'_i)$
    $L'_{\text{MSE}}[i] = \sum(X'_i - \hat{X}'_i)^2$
    Update LSTM-AE weights to minimize L'_{MSE}
end

`/* Phase 4: Threshold setting */`
$\eta = \max(L_{\text{MSE}}, L'_{\text{MSE}})$

`/* Phase 5: Anomaly Detection (Online/Testing) */`
for $i = 0$ **to** m **do**
    $L''_{\text{MSE}}[i] = \sum(X''_i - \hat{X}''_i)^2$
    if $L''_{MSE}[i] > \eta$ **then**
        $X''_i \rightarrow$ Anomaly
    end
end

After the initial training, we fine-tune the pre-trained LSTM-AE using a few days of nominal samples. During this process, the model's parameters are optimized via backpropagation to minimize the reconstruction error, measured in MSE, on the few samples from the target domain, i.e., few-shot.

The purpose of applying our few-shot LSTM-AE technique is to reconstruct the normal samples as accurately as possible on the target domain. During the testing phase, if an anomalous sample is fed into a model that is trained exclusively on normal data, the decoder will attempt to project the encoded representation of this sample into the learned normal data space and reconstruct it as "normal", so the reconstruction will likely be very different from the original input. Thus there will be a large reconstruction error for the anomalous sample, which can then be used as a signal to detect the anomaly.

3.2 Anomaly Detection

Since an anomaly is characterized as an observation that significantly differs from the nominal patterns, a threshold can be set to determine the extent of deviation. Any samples exceeding that threshold are classified as anomalies. Given that we already know the ground truth labels of samples from the training and fine-tuning datasets are nominal, any reconstruction loss for these instances should be considered within the acceptable range. Thus, we can set the threshold η to be the largest of these reconstruction errors. In the last testing phase, test input that has a reconstruction error exceeding the threshold will be classified as an anomaly.

4. CASE STUDY

This study utilizes the MZVAV-2 dataset from the ASHRAE 1312 project, released by Lawrence Berkeley National Laboratory (LBNL) in 2019 for automated fault detection in HVAC systems (Lin, 2019). The dataset contains real and simulated data from a multi-zone air handling unit in an Iowa building, recorded every minute from sensors across four zones. It includes both normal operation and instances of faults, such as a leaking heating coil valve and a stuck cooling coil valve. The real dataset consists of 11 normal days and 2 days with faults, while the simulated dataset contains 13 normal days and 13 days of simulated faults. Each day contains 1,440 data examples, resulting in a total of 18,720 examples for the real dataset and 37,440 examples for the simulated dataset. Both datasets have the same 17 features and one ground truth label column, as summarized in Table 2. The data was captured on different days throughout the year to account for seasonal variations.

4.1 Data Preparation

The simulated dataset, containing a higher proportion of faulty examples, is better suited for showcasing each model's ability to detect different anomaly types. Therefore, the real dataset was used for training, while the simulated dataset was used for fine-tuning and testing.

During the data preparation stage, 11 normal days from the real dataset, totaling 15,840 examples, were used

Table 2. MZVAV-2 Variables Measured. CS is Control Signal.

Variable	Unit	Variable	Unit
Supply Air Temp.	$^\circ F$	Exhaust Air Damper CS	[0,1]
SAT Set Pt.	$^\circ F$	Outdoor Air Damper CS	[0,1]
Outdoor Air Temp.	$^\circ F$	Return Air Damper CS	[0,1]
Mixed Air Temp.	$^\circ F$	Cooling Coil Valve CS	[0,1]
Return Air Temp.	$^\circ F$	Heating Coil Valve CS	[0,1]
Supply Air Fan Status	0/1	Supply Air Duct Press.	psi
Return Air Fan Status	0/1	SAD Static Press. Set Pt.	psi
SAF Speed CS	[0,1]	Occupancy Indicator	0/1
RAF Speed CS	[0,1]	Fault Ground Truth	0/1

to train the models. k days of normal instances were randomly selected from the simulated dataset for fine-tuning. Four normal days and all anomalous days from the simulated dataset, totaling 24,480 examples, were reserved for testing. The real dataset was cleaned and scaled, and the fine-tuning and test datasets were then scaled to match the scale of the training set. Finally, all three sets were reshaped into time sequences. The $Timestep$ was set to 10, meaning that for each data point, the LSTM-AE model considers the current point along with the previous 9 data points. This value was chosen through a hyperparameter search to provide sufficient temporal context for the model to predict the current data point's behavior without incurring excessive computational cost.

4.2 Model Training Details

The experiment setup is described in Algorithm 3.1. The architecture and parameters for the LSTM-AE were determined using a random search of 10 trials. The hyperparameters searched include the number of LSTM layers, the number of units in each layer, and the dropout rate. The number of layers varied from 2 to 6 for both the encoder and decoder, units per layer ranged from 4 to 16 with increments of 4, and the dropout rate was between 0.1 to 0.5 with increments of 0.1. These values were randomly selected within the defined ranges across trials to minimize the validation loss. The LSTM-AE model consists of an encoder and a decoder. The encoder aims to compress the input time series data into a batch of feature sequences with temporal context, constructing a lower-dimensional representation of the input that captures the essential temporal features and patterns within the data. In this architecture, the 17-dimensional input is first fed into an LSTM layer that learns an 8-dimensional output, followed by a 20% dropout layer to prevent overfitting. Another LSTM layer and a dropout layer further reduce the dimensionality to four. The decoder then reconstructs the input sequence from its encoded representation in the latent space. The decoder structure is symmetrical to that of the encoder, inversely mapping the 4-dimensional compressed feature representation to an output of 17 dimensions, matching the original input. The final layer is a TimeDistributed Dense layer, which projects the LSTM output back to the original input shape. The LSTM-AE uses the RELU activation function and is trained with the Adam optimizer and a learning rate of 0.0001. The training of the LSTM-AE is conducted for 100 epochs, and the fine-tuning process requires an additional 10 epochs on the few-shot data.

4.3 Model Evaluation

The performance of our model was evaluated using True Positive Rate (TPR) and False Positive Rate (FPR). A Receiver Operating Characteristic (ROC) Curve was generated to show the tradeoff between TPR and FPR. The area under the ROC curve is known as the Area Under Curve (Curve), which we also used as an evaluation metric. The formula for these metrics is shown below:

$$TPR(\text{True Positive Rate/Recall}) = \frac{TP}{TP + FN} \tag{1}$$

$$FPR(\text{False Positive Rate}) = \frac{FP}{FP + TN} \tag{2}$$

$$AUC_{\text{ROC}} = \int_0^1 \text{TPR}\, d\text{FPR} \tag{3}$$

We compared the proposed method with four unsupervised methods proposed by Coursey et al. (2023) that perform anomaly detection on the same dataset: an LSTM-AE (AE) with the same hyperparameter grid-search and training but without few-shot fine-tuning, and three clustering-based autoencoders assigning clusters based on Euclidean distance (EUC CT-AE), Nearest Neighbor (1NN CT-AE), and an Artificial Neural Network (ANN CT-AE). We trained and evaluated each method 25 times. In our few-shot method, we randomly selected k days of target data to fine-tune for each trial. This random selection process was repeated for each value of k within the range [1, 2, 3, 4], ensuring our predictions were not influenced by a biased selection of the few-shot set.

4.4 Discussion of Results

Table 3 summarizes the performance of each method with differing numbers of days used to fine-tune. Notably, the few-shot model when $k = 3$ achieved the highest AUC of 0.873 and the best TPR of 0.809, indicating that our model was effective in identifying actual anomalies or faults in the building system. This is crucial for early detection of system issues, allowing for timely maintenance and repairs. The ROC curve in Figure 1 also shows that few-shot models generally have larger AUC, which is an indication that our method is better at distinguishing between anomalies and normal points.

To investigate whether the introduction of additional nominal samples for few-shot fine-tuning enhances AUC, we perform pairwise one-sided t-tests between models with different k values. The result, summarized in Table 4 reveals a statistically significant improvement in AUC when moving from k =1 to k =3, as well as from k =2 to k =3. Additionally, despite the mean AUC being lower for k =4, the transition from k =3 to k =4 does not show a statistically significant change. These findings demonstrate that generally, few-shot performs equal or better with an increase in the number of few-shot examples considered for fine-tuning k.

However, it is worth noting that our few-shot models have higher FPRs than other methods. For anomaly detection on complex systems with limited data, there is often a tradeoff between TPR and FPR. With more shots are used, the FPR decreases as the model learns to better distinguish normal variations from true anomalies

Table 3. Averaged AUC, FPR, and TPR for 25 of each model type. Each statistic is presented as mean ± standard deviation.

Model	AUC	TPR	FPR
AE	0.773±0.033	0.616±0.070	0.069±0.051
1NN CT-AE	0.840±0.028	0.727±0.079	0.099±0.049
ANN CT-AE	0.839±0.027	0.723±0.075	0.100±0.043
EUC CT-AE	0.834±0.017	0.730±0.038	0.053±0.026
Few-Shot k=1	0.855±0.022	0.789±0.029	0.168±0.058
Few-Shot k=2	0.862±0.013	0.788±0.042	0.137±0.077
Few-Shot k=3	0.873±0.015	0.809±0.014	0.122±0.055
Few-Shot k=4	0.868±0.022	0.803±0.030	0.116±0.051

Table 4. Pairwise comparison of few-shot models' AUC using one-sided t-tests, testing whether Model 2 has a significantly higher AUC than Model 1. Significance levels are indicated as * (p <0.05) and ** (p <0.01).

Model 1	Model 2	t-statistic	p-value
Few-Shot 1	Few-Shot 2	-1.069	0.147
Few-Shot 1	Few-Shot 3	-2.552	0.008**
Few-Shot 1	Few-Shot 4	-1.588	0.062
Few-Shot 2	Few-Shot 3	-2.118	0.022*
Few-Shot 2	Few-Shot 4	-0.886	0.192
Few-Shot 3	Few-Shot 4	0.722	0.762

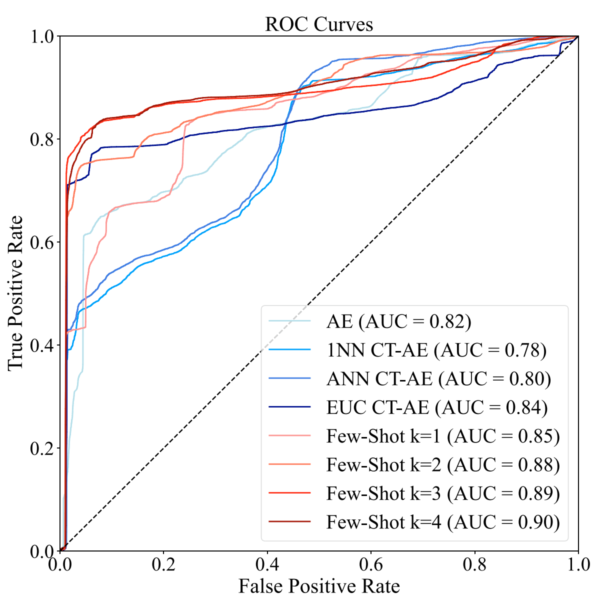

Fig. 2. ROC Curves of all models. This is only for one instance of a trained model.

with increased data. Despite false alarms can lead to unnecessary maintenance interventions, operators can use the duration of an alarm as an additional criterion to assess the model's credibility. Genuine faults in HVAC systems often persist for a relatively long time, ranging from several minutes to hours or even days, while false alarms caused by sensor noise are typically short-lived.

4.5 Anomaly Categories

The LBNL dataset also labels the categories of each anomaly, and each fault type is injected into the system

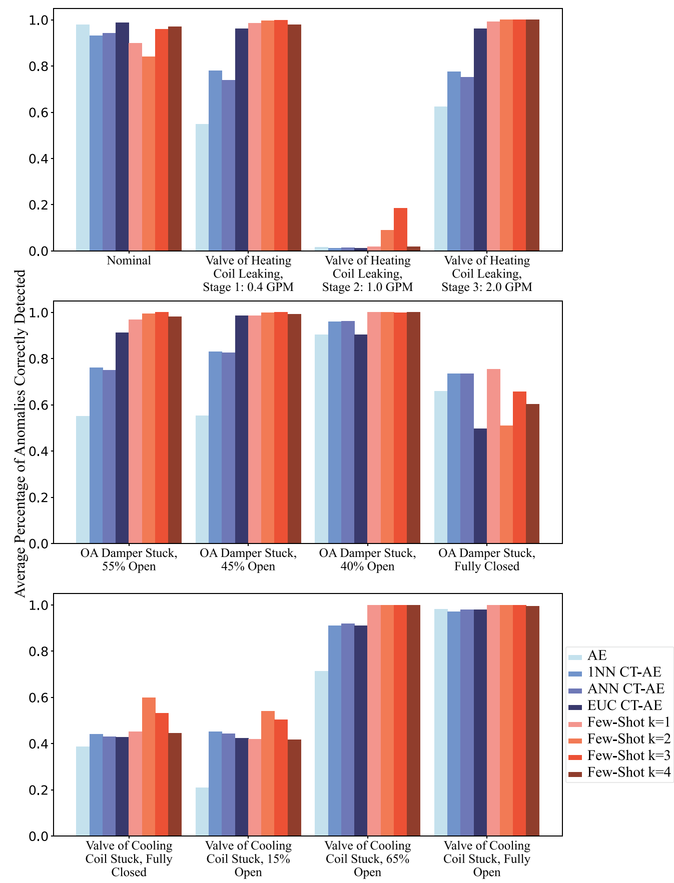

Fig. 3. Average percentage of accuracy of each model's ability to detect anomalies in each fault category

for an entire day. The average accuracy for predicting each type across 25 trials is displayed in Figure 3. As can be seen, few-shot demonstrated a significant advantage in detecting all 11 types of faults. However, the enhanced sensitivity of few-shot models came at the cost of a higher false positive rate, as the prediction accuracy for nominal data was generally lower than the other methods. Despite that, the prediction accuracy for nominal data improved by increasing the number of shots, which provides further evidence that the model benefits from a greater number of examples to learn normal patterns in the target domain.

It is noteworthy that none of the methods perform very well on the Valve of the Heating Coil Leaking at 1 GPM, the OA Damper Stuck Fully Closed, and the Valve of the Cooling Coil Stuck (Fully Closed and 15% Open). This could be attributed to the similar operational patterns that these faults share with normal functioning conditions, which is hard for all models to distinguish. Particularly, the Valve of the Heating Coil Leaking at a rate of 1.0 GPM closely resembles the rate of water flow through the valve under normal conditions, so it is challenging for the models to recognize it as an anomaly without additional data or knowledge.

5. CONCLUSION

In this paper, we presented a domain adaptation-based few-shot anomaly detection method for HVAC systems that uses LSTM autoencoders. Our approach improves the accuracy of anomaly detection compared to unsupervised methods with just a few labeled data instances. We demonstrate that with the capacity to leverage a small set of normal samples from the target domain for fine-tuning, an existing trained model can detect a number of different fault types with higher accuracy than a variety of state-of-the-art clustering-based autoencoder methods for anomaly detection.

One limitation of our method is that it relies solely on normal samples during the few-shot domain adaptation process. This approach assumes a consistent definition of 'normal' operating modes across different systems, a premise that may not always be valid. As a result, the model may be predisposed to higher false positive rates, especially in scenarios where the anomalous behavior deviates subtly from the 'normal' patterns it has been trained to recognize. To address the problem of limited amount of target domain data, future work could explore the possibilities of iteratively retraining the model with newly acquired labeled data from the target domain and incorporating a select number of anomaly examples for the model to compare between normal samples and true anomalies, thereby helping the model to become more discriminating in the new domain.

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